



AUTONOMOUS GARBAGE-COLLECTING ROBOT WITH CLAW

CSE46I : INTRODUCTION TO ROBOTICS

GROUP 5 | SECTION 6

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ABSTRACT

The Autonomous Garbage-Collecting Robot with Claw is a low-cost robotic system that will be able to collect small pieces of litter in indoor and semi-indoor environments. The robot is controlled by an Arduino Uno and has two HC-SR04 ultrasonic sensors for wall detection and serpentine lane navigation. Target garbage objects are detected by the IR/ESP32 sensor, and then the robot stops, and the mechanical arm composed of two servos is extended to pick up the garbage and transfer it to the garbage container. The movement is provided by four DC motors that are connected to an L298N motor driver. Weight of the container is monitored by a load cell with the help of HX711 and a buzzer is triggered when the weight reaches. The goal is to lessen the need to collect litter manually, reduce the contact with unhygienic waste, and offer an easy and cheap robotic clean-up solution.

INTRODUCTION

Collecting garbage in communal areas is repetitive, time consuming and unhygienic. The aim of this project is to design an autonomous robot which can move in a controlled area, identify small garbage objects and pick up the garbage with a mechanical claw and store it in an in-board container.

Objectives:

- Navigate autonomously using dual HC-SR04 sensors
- Detect garbage using an ESP32 Cam/IR sensor
- Pick and drop waste using a servo-controlled claw
- Monitor container weight using a load cell and HX711 and alert users with a buzzer

APPLICATION & COMPETITORS

Applications:

- Self-service litter pick-up in classrooms, corridors, offices and campuses.
- Light waste collection including paper, wrappers and bottle caps.
- Saves manual cleaning time and effort.

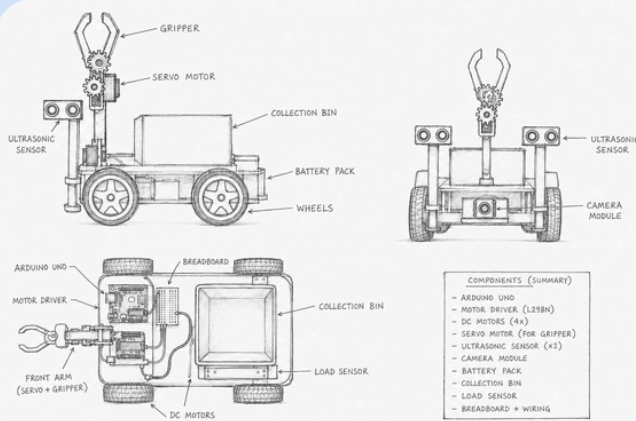
Competitor Comparison:

Feature	Our Robot	Existing Robot
Operation	Autonomous	Autonomous/Remote
Detection	IR + Ultrasonic sensors	Advanced sensors / camera
Cost	Low	Expensive

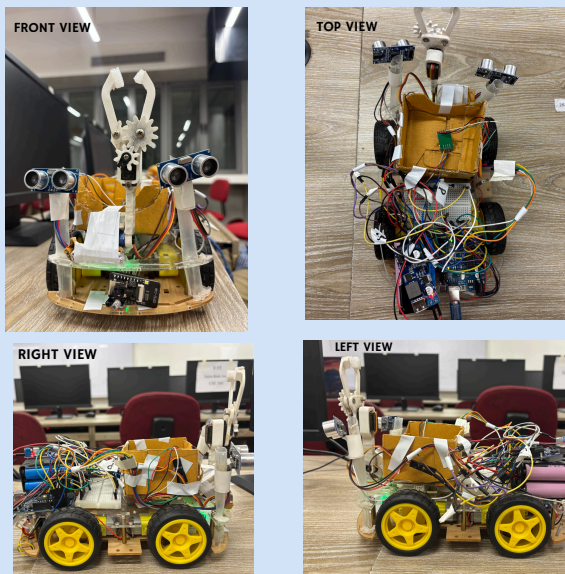
Advantages: Self-powered, inexpensive, replaceable sensors/parts.

Limitations: Suitable for light garbage, and suitable for controlled environments, cannot distinguish the type of garbage

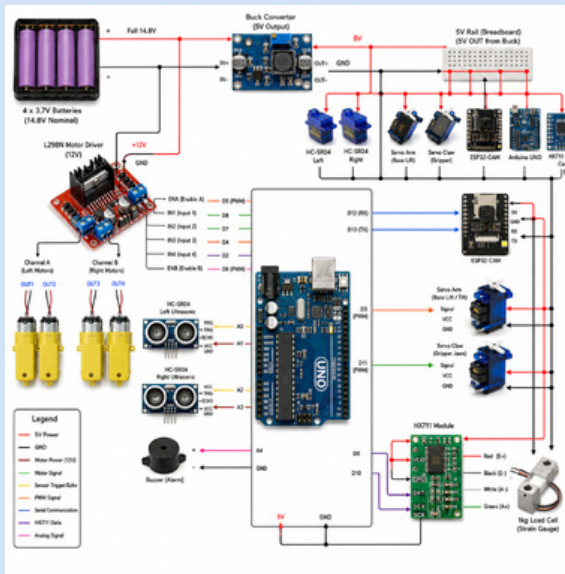
SKETCH/CONCEPT DIAGRAM



STRUCTURAL DIAGRAM



CONTROL CIRCUIT DIAGRAM



HARDWARE COMPONENTS

Arduino Uno	Main Controller
HC-SR04 ×2	detect wall and obstacles
IR Sensor/ESP sensor	Detects garbage objects
L298N Motor Driver	moves the motor
DC Gear Motors ×4	drives the motor
SG90 Servo Motors ×2	control arm and claw
Load Cell + HX711	weight of garbage
Buzzer	to warn bin is full
Battery Supply and LM2596S Buck Converter	regulate power
4WD Chassis	robot structure

FUNCTIONALITIES

Mechanical Functionality

- 4-wheel drive using four DC motors.
- Two SG90 servos control the arm and claw.
- Claw picks up and drops garbage.

Software Functionality

- Arduino Uno runs the control logic.
- Controls movement, pickup, and bin monitoring.
- Weight threshold triggers the buzzer.

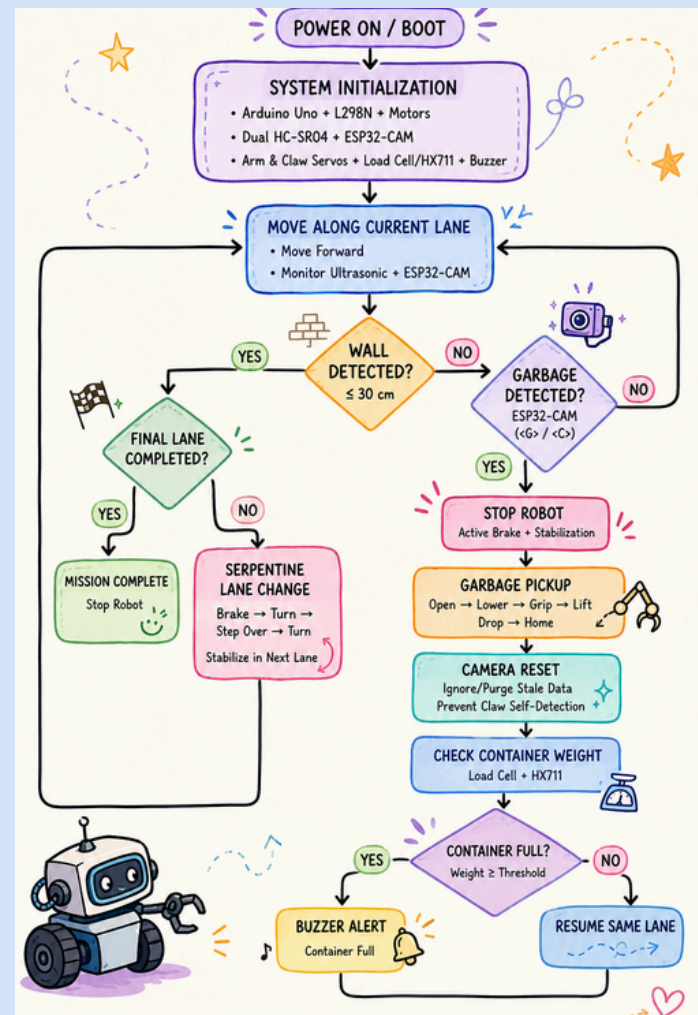
Navigation Functionality

- Serpentine lane-based navigation.
- Dual HC-SR04 sensors detect obstacles.
- Turn → Step Over → Turn between lanes.

Software-Hardware Communication

- UART: ESP32-CAM, ↔ Arduino serial communication
- Data flow: Sensors → Arduino → Motors/Servos/Buzzer

WORKFLOW DIAGRAM



CONCLUSION

The proposed robot combines autonomous navigation, garbage detection, mechanical pickup, and bin-weight monitoring in a low-cost robotic platform. The system demonstrates lane-based movement, wall detection using dual ultrasonic sensors, claw-based collection of lightweight garbage, and full-bin monitoring. Key Achievements: Successful integration of navigation, sensing, motor control, claw operation, and container monitoring. Future Improvements: Camera-based waste classification, improved path planning, a stronger claw mechanism, and a rechargeable power system.